

1) i

Day	A(x)	B(y)	x^2	y^2	xy
1	50	45	2500	2025	2250
2	60	55	3600	3025	3300
3	45	50	2025	2500	2250
4	55	60	3025	3600	3300
5	70	65	4900	4225	4550
6	65	70	4225	4900	4550

$$\sum x = 345 \quad \sum y = 345 \quad \sum x^2 = 20275 \quad \sum y^2 = 20275$$

$$\sum xy = 20250 \quad n = 6$$

$$r = \frac{n \sum xy - (\sum x)(\sum y)}{\sqrt{[n \sum x^2 - (\sum x)^2][n \sum y^2 - (\sum y)^2]}}$$

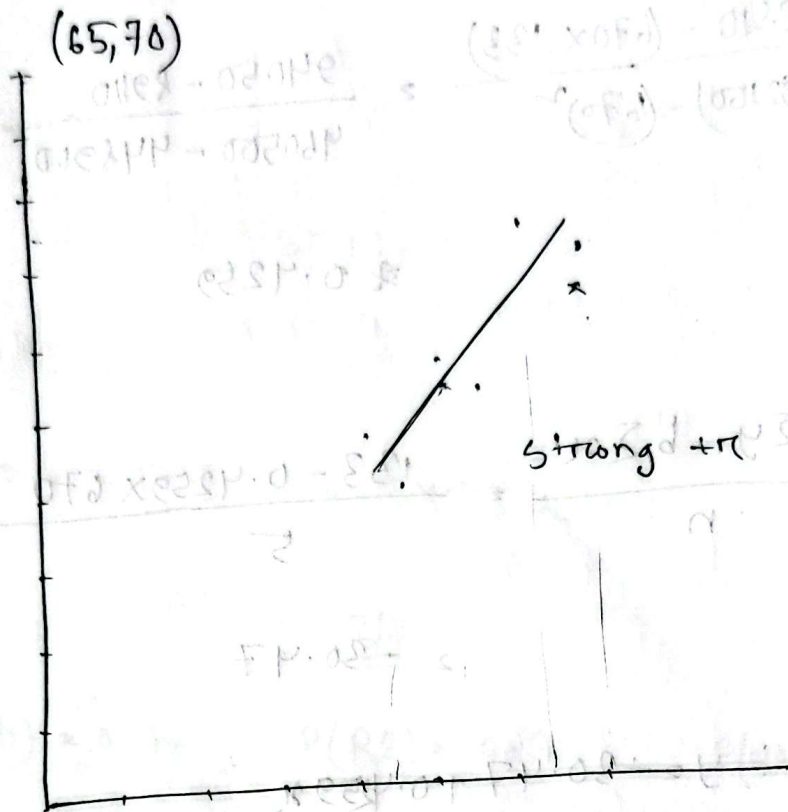
$$= \frac{6 \times 20250 - (345 \times 345)}{\sqrt{[6 \times 20275 - 345^2][6 \times 20275 - 345^2]}}$$

$$= \frac{2175}{2625} \approx 0.8286 \approx 0.83$$

there's a strong correlation between the A & B

ii)

(x,y) Points $\rightarrow (50,45), (60,55), (45,50), (55,60), (70,65)$



2/i

Regression line of y on x:

$$x = [150, 120, 140, 150, 160]$$

$$y = [12, 22, 28, 31, 40]$$

x	y	x^2	xy
150	12	15000	1200
120	22	14400	2640
140	28	19600	3960
150	31	22500	4650
160	40	25600	6400

$$\sum x = 670 \quad \sum y = 133$$

$$\sum x^2 = 92100 \quad \sum xy = 18810$$

$$b_2 = \frac{n \sum xy - \sum x \sum y}{n \sum x^2 - (\sum x)^2}$$

$$= \frac{5 \times 18810 - (670 \times 133)}{(5 \times 92100) - (670)^2} = \frac{94050 - 89110}{460500 - 448900}$$

$$= 0.4259$$

$$y = a + bx$$

$$\therefore a = \frac{\sum y - b \sum x}{n} = \frac{133 - 0.4259 \times 670}{5}$$

$$= -30.47$$

$$\therefore y = -30.47 + 0.4259x$$

$$ii) y = -30.47 + 0.4259 \times 180 = 45.19$$

iii)

$$y = a + by$$

$$\sum y^2 = 144 + 22^2 + 28^2 + 31^2 + 40^2 = 3973$$

$$b_2 = \frac{n \sum xy - \sum x \sum y}{n \sum y^2 - (\sum y)^2} = \frac{5 \times 18810 - 670 \times 133}{5 \times 3973 - (133)^2}$$

$$= \frac{5 \times 18810 - 670 \times 133}{5 \times 3973 - (133)^2} = \frac{94050 - 89110}{19865 - 17689} = \frac{4940}{2176} = 2.27$$

$$x = a + by$$

$$\text{on, } a = \frac{\sum x - b \sum y}{n} = \frac{670 - 17.96 \times 133}{5} = -343.30$$

$$x = -343.30 + 17.96y$$

$$\text{iv) } y = 45$$

$$x = -343.30 + 17.96 \times 45 \approx 465$$

$$P(R_1) = 0.4, P(R_2) = 0.35, P(R_3) = 0.25$$

$$P(L/R_1) = 0.005, P(L/R_2) = 0.01, P(L/R_3) = 0.015$$

Find $P(R_2/L)$, Find $P(R_3/L)$

Total Probability of Loss:

$$P(L) = (0.4 \times 0.005) + (0.35 \times 0.01) + (0.25 \times 0.015) = 0.00925$$

$$\text{i) } P(R_2/L) = \frac{P(R_2) \cdot P(L/R_2)}{P(L)} = \frac{0.35 \times 0.01}{0.00925} \approx 0.378$$

$$\text{ii) } P(R_3/L) = \frac{P(R_3) \cdot P(L/R_3)}{P(L)} = \frac{0.25 \times 0.015}{0.00925} \approx 0.405$$

4)

$$P(A) = 0.3$$

$$P(B) = 0.45$$

$$P(C) = 0.25$$

$$P(F/A) = 0.01$$

$$P(F/B) = 0.02$$

$$P(F/C) = 0.03$$

Total ~~fail~~ of probability of fail.

$$P(F) = (0.3 \times 0.01) + (0.45 \times 0.02) + (0.25 \times 0.03)$$

$$= 0.0195$$

$$i) P(A/F) = \frac{P(A) \cdot P(F/A)}{P(F)} = \frac{0.3 \times 0.01}{0.0195} \approx 0.154$$

$$ii) P(C/F) = \frac{P(C) \cdot P(F/C)}{P(F)} = \frac{0.25 \times 0.03}{0.0195} \approx 0.385$$

(5)

~~if $x=4$,~~ $f(x) = \frac{1}{24}(x^2+1); 1 < x < 4$

$$\int_1^4 \frac{1}{24}(x^2+1) dx = \frac{1}{24} \int_1^4 (x^2+1) dx = \frac{1}{24} \left[\frac{x^3}{3} + x \right]_1^4$$

$$= \frac{1}{24} \left\{ \left(\frac{64}{3} + 4 \right) - \left(\frac{1}{3} + 1 \right) \right\}$$

$$= \frac{1}{24} \cdot \frac{72}{3} = \frac{1}{24} \cdot 24 = 1$$

$$ii) E(X) = \int_1^4 x \cdot f(x) dx = \frac{1}{24} \int_1^4 x(x^2+1) dx$$

$$= \frac{1}{24} \int_1^4 (x^3+x) dx = \frac{1}{24} \left[\frac{x^4}{4} + \frac{x^2}{2} \right]_1^4$$

$$= \frac{1}{24} \left\{ (64+8) - (0.25+0.5) \right\} = \frac{1}{24} \cdot 71.25 = 2.96875$$

$$iii) SD(X)$$

$$E(X^2) = \int_1^4 x^2 \cdot f(x) dx = \frac{1}{24} \int_1^4 x^2(x^2+1) dx$$

$$= \frac{1}{24} \int_1^4 (x^4+x^2) dx = \frac{1}{24} \left[\frac{x^5}{5} + \frac{x^3}{3} \right]_1^4$$

$$= \frac{1}{24} \left(\frac{1024}{5} + \frac{64}{3} - \left(\frac{1}{5} + \frac{1}{3} \right) \right) = \frac{1}{24} (204.8 + 21.33 - 0.533)$$

$$= 9.4$$

$$\text{Var}(X) = E(X^2) - (E(X))^2$$

$$= 9.4 - (2.97)^2 \approx 0.586$$

$$\text{SD}(X) = \sqrt{0.586} \approx 0.77$$

$$\text{iii) } P(X < 3.5) = \int_1^{3.5} f(x) dx = \frac{1}{24} \left[\frac{x^3}{3} + x \right]_1^{3.5}$$

$$= \frac{1}{24} \left(\frac{42.875}{3} + 3.5 - \frac{1}{3} - 1 \right)$$

$$\approx 0.686$$

$$\text{iv) } P(X > 0.3)$$

Since 0.3 is outside the support $[1, 4]$, the PDF is zero below $x=1$, so,

$$P(X > 0.3) = 1$$

(6)
 $f(x) = k(x-1); 2 \leq x \leq 6$

i) $\int_2^6 k(x-1) dx = 1$

or, $k \int_2^6 (x-1) dx = 1$

Now,

~~$\int_2^6 (x-1) dx = \int_2^6 \left(\frac{x^2}{2} - x \right) dx$~~

~~$\left[\frac{x^2}{2} - x \right]_2^6 = \frac{(6)^2}{2} - 6$~~

$\int_2^6 (x-1) dx = \int_2^6 \left[\frac{(x-1)^2}{2} \right] dx$

$\left[\frac{(x-1)^3}{3} \right]_2^6 = \frac{1}{3} \left[\frac{(x-1)^3}{2} \right]_2^6 = \frac{25}{12} - \frac{1}{2} = 12$

$\therefore k \cdot 12 = 1$

or, $k = \frac{1}{12}$

$$\begin{aligned} \text{ii) } E(X) &= \int_2^6 x f(x) dx = \frac{1}{12} \int_2^6 x(x-1) dx \\ &= \frac{1}{12} \int_2^6 (x^2 - x) dx = \frac{1}{12} \left[\frac{x^3}{3} - \frac{x^2}{2} \right]_2^6 \\ &= \frac{1}{12} \left\{ \left(\frac{6^3}{3} - \frac{6^2}{2} \right) - \left(\frac{2^3}{3} - \frac{2^2}{2} \right) \right\} \end{aligned}$$

$$= \frac{1}{12} \cdot \frac{160}{3} = 4.44$$

$$\text{iii) } CV(X) = \frac{\sigma}{\mu} \quad \mu = E(X) = 4.44$$

$$\begin{aligned} E(X^2) &= \int_2^6 x^2 f(x) dx = \frac{1}{12} \int_2^6 x^2(x-1) dx \\ &= \frac{1}{12} \int_2^6 (x^3 - x^2) dx \end{aligned}$$

$$\begin{aligned} &= \frac{1}{12} \left[\frac{x^4}{4} - \frac{x^3}{3} \right]_2^6 = \frac{1}{12} \left[\left(\frac{6^4}{4} - \frac{6^3}{3} \right) - \left(\frac{2^4}{4} - \frac{2^3}{3} \right) \right] \\ &= \frac{188}{9} \end{aligned}$$

$$\text{Var}(X) = E(X^2) - (E(X))^2 = \frac{188}{9} - \left(\frac{40}{9} \right)^2$$

$$= \frac{188}{9} - \frac{1600}{81} = \frac{92}{81}$$

$$\sigma = \sqrt{\frac{92}{81}} = \frac{\sqrt{92}}{9}$$

$$CV(X) = \frac{\sigma}{\mu} = \frac{\frac{\sqrt{92}}{9}}{\frac{40}{9}} = \frac{\sqrt{92}}{40} \approx 0.2398$$

(7)

$$f(x) = \frac{3}{12} (6x - 3x^2) = \frac{1}{4} (6x - 3x^2) = \frac{3}{2} x - \frac{3}{4} x^2$$

$$f(x) = \frac{3}{2} x - \frac{3}{4} x^2; 0 < x < 2$$

Check if it is a valid PDF.

$$\int_0^2 f(x) dx = \int_0^2 \left(\frac{3}{2} x - \frac{3}{4} x^2 \right) dx$$

$$= \left[\frac{3}{4} x^2 - \frac{3}{12} x^3 \right]_0^2 = \left(\frac{3}{4} \cdot 4 - \frac{3}{12} \cdot 8 \right) = (3 - 2) = 1$$

(valid)

$$i) P(X \leq 1.5) = \int_0^{1.5} f(x) dx = \int_0^{1.5} \left(\frac{3}{2} x - \frac{3}{4} x^2 \right) dx$$

$$= \left[\frac{3}{4} x^2 - \frac{3}{12} x^3 \right]_0^{1.5} = \frac{3}{4} (1.5)^2 - \frac{3}{12} (1.5)^3$$

$$= \frac{3}{4} (2.25) - \frac{3}{12} (3.375) = 1.6875 - 0.84375 = 0.84375$$

$$ii) P(0.5 < x < 1.5) = \int_{0.5}^{1.5} f(x) dx = \left[\frac{3}{4} x^2 - \frac{3}{12} x^3 \right]_{0.5}^{1.5}$$

$$= \left[\left(\frac{3 \times (1.5)^2}{4} - \frac{3 \times (1.5)^3}{12} \right) - \left(\frac{3 \times (0.5)^2}{4} - \frac{3 \times (0.5)^3}{12} \right) \right]$$

$$= 0.84375 - 0.15625$$

$$= 0.6875$$

iii) Mean and Variance

$$\text{Mean: } E(x) = \int_0^2 x f(x) dx$$

$$= \int_0^2 x \left(\frac{3}{2} x - \frac{3}{4} x^2 \right) dx$$

$$= \int_0^2 \left(\frac{3x^2}{2} - \frac{3x^3}{4} \right) dx$$

$$= \left[\frac{3x^3}{6} - \frac{3x^4}{16} \right]_0^2 = \left[\frac{1}{2} x^3 - \frac{3}{16} x^4 \right]_0^2$$

$$= \frac{2^3}{2} - \frac{3 \times 2^4}{16} = 1$$

$$\text{Variance: } E(X^2) = \int_0^2 x^2 f(x) dx$$

$$= \int_0^2 x^2 \left(\frac{3}{2}x - \frac{3}{4}x^2 \right) dx$$

$$= \int_0^2 \left(\frac{3x^3}{2} - \frac{3x^4}{4} \right) dx$$

$$= \left[\frac{3}{2} \frac{x^4}{4} - \frac{3}{4} \frac{x^5}{5} \right]_0^2$$

$$= \left[\frac{3x^4}{8} - \frac{3x^5}{20} \right]_0^2$$

$$= \left[\frac{3 \times (2)^4}{8} - \frac{3 \times (2)^5}{20} \right]$$

$$= 1.2$$

$$\text{Var}(X) = E(X^2) - (E(X))^2 = 1.2 - 1^2 = 0.2$$

Average rate of login failures per hour

$$\lambda = 5$$

The poisson probability mass function is:

$$P(x) = \frac{e^{-\lambda} \lambda^x}{x!}$$

i)

$$k = 4$$

$$\lambda = 5$$

$$P(x) = \frac{e^{-5} 5^4}{4!} = \frac{e^{-5} 625}{24} = \frac{0.00673 \times 625}{24} = 0.17546$$

$$ii) P(x \leq 2) = P(0) + P(1) + P(2)$$

$$P(0) = \frac{e^{-\lambda} \lambda^0}{0!} = \frac{e^{-5} 5^0}{1} = 0.006737$$

$$P(1) = \frac{e^{-\lambda} \lambda^1}{1!} = \frac{e^{-5} 5^1}{1} = 0.03369$$

$$P(2) = \frac{e^{-5} 5^2}{2!} = 0.08422$$

$$P(X \leq 2) = 0.006737 + 0.03369 + 0.08422 = 0.12465$$

$$\text{ii)} P(X \geq 3) = 1 - P(X \leq 2)$$

$$P(X \geq 3) = 1 - 0.1247 = 0.8753$$

(9)

	Python	Java	Row Total ↓
Morning	40 E_{11}	15 E_{12}	55
Evening	25 E_{21}	30 E_{22}	55
Column → Total	65	45	110 ↘ Grand Total

$$E_{ij} = \frac{(\text{Row Total}) \times (\text{Column Total})}{\text{Grand Total}}$$

$$E_{11} = \frac{55 \times 65}{110} = 32.5$$

$$E_{12} = \frac{55 \times 45}{110} = 22.5$$

$$E_{22} = \frac{55 \times 45}{110} = 22.5$$

$$E_{21} = \frac{55 \times 65}{110} = 32.5$$

Chi-Square Test:

$$\chi^2 = \sum \frac{(O - E)^2}{E}$$

O → Actual number
from the ques
Data

$$\therefore \chi^2 = \frac{40 - 32.5}{32.5}$$

$$\chi^2 = \frac{(O - E_{11})^2}{E_{11}} + \frac{(O - E_{12})^2}{E_{12}} + \frac{(O - E_{21})^2}{E_{21}} + \frac{(O - E_{22})^2}{E_{22}}$$

$$= \frac{(40 - 32.5)^2}{32.5} + \frac{(15 - 22.5)^2}{22.5} + \frac{(25 - 32.5)^2}{32.5} + \frac{(30 - 22.5)^2}{22.5}$$

$$\chi^2 = 8.4616$$

Degrees of Freedom = (Total rows) × (Total columns)

$$(r - 1)(c - 1) = (2 - 1)(2 - 1)$$

$$= 1 \times 1 = 1$$

r → row
c → column

Tabulated chi-square value at 5% significance level and 1df: 3.84

Since.

$$\chi^2_{\text{calc}} = 8.46 > 3.84 = \chi^2_{\text{tabulated}}$$

There is a significant association between the programming language used and the time of day when coding errors occur.